

# Md Samshad Rahman

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## Professional summary:

**Data & Applied ML Engineer** with dual Master's degrees in Applied Computer Science and Statistics and 4+ years of experience architecting end-to-end data pipelines. Specializes in building automated ETL workflows, statistical anomaly detection systems, and executive dashboards that cut operational overhead and accelerate leadership decisions. Bridges containerized backend services with applied ML and LLM pipelines to deliver resilient, high-throughput data architecture.

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## Technical Skills:

**Engineering Practices:** Agile Methodologies, Code Quality, Refactoring, Unit Testing, Integration Testing, System Reliability, Software Security, TDD (Test-Driven Development), Automated Testing.

**Analytics & Reporting:** Google Analytics, Dash/Plotly, Looker Studio, Tableau, Advanced Excel, KPI & Performance Metrics, Process Mapping, Business Process Re-engineering, Data Validation & QA, Hypothesis-Driven Analysis.

**Machine Learning & AI:** LLM Fine-Tuning (Llama, LoRA, Unsloth), PyTorch, Hugging Face, Scikit-Learn, spaCy, Ollama, Claude, Anomaly Detection, Clustering, Anthropic Claude API.

**DevOps & Tools:** Docker, Git, CI/CD (GitHub Actions), AWS (EC2, S3, Lambda), GCP, Jira, Confluence, CloudFormation, Kubernetes, Microservices Architecture, OpenAPI, Swagger.

**Backend:** Python (Pandas, NumPy, Pydantic, PyTest), SQL (PostgreSQL, MySQL), FastAPI, Flask, REST APIs, Redis, Celery, Taskiq, Data Pipelines, NGINX.

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## Professional Experience:

**Dalhousie University**, Halifax, Nova Scotia

**Data Specialist & Graduate Researcher (*Persuasive Computing Lab*)** [September 2024 – April 2025]

- Cut public procurement research time by 60% by replacing a legacy Nova Scotia government reporting workflow with automated Python ETL pipelines, standardizing 10 years of raw records and building an interactive Dash tool that removed 35% of dataset noise.
- Built the FastAPI backend and React Native integration for an AI eating-coach app, using personality-trait detection (Big Five) to personalize chat guidance in real time.
- Fine-tuned and deployed an open-source Llama 3.2 (3B) model on T4 GPU infrastructure using FastAPI, establishing 1,500 verified test cases to eliminate response hallucinations.

**Systems Analyst (*Data Integrity & Billing*)**, Munshi Engineers, Bangladesh [March 2020 – April 2023]

- Built automated SQL and Python validation scripts across two national utility systems (DPDC, BPDB) that flagged consumption anomalies before monthly invoice runs, preventing billing errors across high-volume customer accounts.
- Turned raw operational data and transaction data into automated SQL and Power BI reports, leading a 4-person team through the process and helping leadership monitor operational bottlenecks and increase revenue.
- Replaced a resource-heavy legacy prototype with an event-driven AWS Lambda pipeline, cutting infrastructure costs by approximately 50% while lifting processing throughput by over 30%.
- Wrote data dictionaries and validation protocols for national infrastructure datasets, keeping records audit-ready and data consistent across teams.

**Python Developer (Contract)**, Smartbytes Ltd., Bangladesh [January 2020 – April 2020]

- Built an NLP ingestion pipeline (spaCy, LinearSVC, BeautifulSoup, Selenium) that parsed over 4 million unstructured documents into normalized PostgreSQL schemas, lifting matching accuracy from 65% to 92%.
- Containerized and deployed the extraction engine as a FastAPI microservice on AWS EC2 with auto-generated OpenAPI/Swagger documentation, streamlining frontend integration.

**Junior Software Engineer (Contract)**, Free Pixel Games Limited [September 2019 – November 2019]

- Standardized data from 5+ external providers into one normalized MySQL warehouse, resolving schema conflicts across ~50,000 monthly records with zero downtime.
- Automated the daily revenue reconciliation workflow, cutting manual consolidation time by over 80% (2 hours down to 15 minutes).

### **Additional Experience:**

**Microsoft Senior Advisor**, Best Buy Canada Ltd. Halifax, NS [October 2023 – Present]

- Deliver technical advisory, hardware diagnostics, and Microsoft 365 environment support to retail and commercial clients.
- Troubleshoot OS, networking, hardware, and peripheral compatibility issues, translating technical requirements into plain-language solutions for end users.

### **Education:**

- **Master of Applied Computer Science** [September 2023 – May 2025]  
*Dalhousie University, Halifax, NS*
- **Master's in Applied Statistics and Data Science** [November 2019 – April 2022]  
*Jahangirnagar University, Bangladesh*
- **BSc. in Computer Science and Software Engineering** [January 2015 – January 2020]  
*American International University-Bangladesh*

### **Volunteer & Leadership Experience:**

**Community Volunteer**, Bangladesh Canada Friendship Society (BCFS). Halifax, NS [2025 – Present]

- Support event execution for a Bangladeshi community organization in Halifax, contributing to concerts, cultural festivals, social gatherings, and sports competitions.
- Actively supported 5+ community events to date, assisting with logistics, setup, and attendee coordination as an on-call volunteer.

**Competitive Programmer and Trainer**, AIUB Competitive Programmers' Community [2017 – 2019]

- Directly trained and mentored 70+ students in data structures, algorithms, and problem-solving techniques.
- Coordinated junior teams across 20+ programming contests, spanning intra-university and national-level competitions.
- Designed and hosted internal contests as a problem setter, while personally competing and representing the university in national inter-university programming contests.